Tema 2

Ex. 1 a) $P(x+y \geq 10 \mid x = par) = \frac{P(x+y \geq 10 \wedge x = par)}{P(x = par)}$

$P(x = par) = \frac{3}{6} \cdot \frac{6}{6} = \frac{3}{6}$

$\left.\begin{matrix} 4+6 \geq 10 \\ 6+4 \geq 10 \\ 6+5 \geq 10 \\ 6+6 \geq 10 \end{matrix}\right\} \Rightarrow P(x+y \geq 10 \wedge x = par) = \frac{4}{36}$

$\Rightarrow P(x+y \geq 10 \mid x = par) = \frac{\frac{4}{36}}{\frac{3}{6}} = \frac{4 \cdot 6}{36 \cdot 3} = \frac{4}{18} = \frac{2}{9}$

b) $P(x+y \geq 10 \mid x \text{ sau } y = par)$

$P(x \text{ sau } y \text{ par}) = P(x = par) + P(y = par) - P(x = par \wedge y = par)$

$= \frac{1}{2} + \frac{1}{2} - \frac{9}{36} = \frac{3}{4}$

$\left.\begin{matrix} 4+6 \geq 10 \\ 5+6 \geq 10 \\ 6+4 \geq 10 \\ 6+5 \geq 10 \\ 6+6 \geq 10 \end{matrix}\right\} \Rightarrow P(x+y \geq 10 \wedge x \text{ sau } y = par) = \frac{5}{36}$

$P(x+y \geq 10 \mid x \text{ sau } y = par) = \frac{\frac{5}{36}}{\frac{3}{4}} = \frac{5 \cdot 4}{36 \cdot 3} = \frac{5}{27}$

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c) $P(X \text{ și } Y \text{ par} \mid X+Y=8)$

$P(X+Y=8) = \frac{5}{36}$

$\{(2,6), (3,5), (4,4), (5,3), (6,2)\}$

$P(X \text{ și } Y \text{ par} \cap X+Y=8) = \frac{3}{36}$

$P(X \text{ și } Y \text{ par} \mid X+Y=8) = \frac{\frac{3}{36}}{\frac{5}{36}} = \frac{3}{5}$

d) $P(X+Y \geq 10) = \frac{6}{36}$

$\{(4,6), (5,5), (5,6), (6,4), (6,5), (6,6)\}$

$P(X = \text{par}) = \frac{1}{2}$

$P(X+Y \geq 10 \cap X = \text{par}) = \frac{4}{36}$

$P(X+Y \geq 10) \cdot P(X = \text{par}) = \frac{6}{36} \cdot \frac{1}{2} = \frac{3}{36} \neq \frac{4}{36} \Rightarrow A \text{ și } B \text{ nu sunt independente}$

Ex. 2/ $P(\text{Spam}) = 0,3$

$P(\text{legitim}) = 0,7$

$P(\text{oferta} \mid \text{Spam}) = 0,8$

$P(\text{oferta} \mid \text{legitim}) = 0,15$

a) $P(\text{Spam} \mid \text{oferta}) = \frac{P(\text{oferta} \mid \text{Spam}) \cdot P(\text{Spam})}{P(\text{oferta})} = \frac{\frac{4}{5} \cdot \frac{3}{10}}{\frac{69}{200}} = \frac{12 \cdot 200}{69 \cdot 50} = \frac{48}{69}$

$P(\text{oferta}) = 0,8 \cdot 0,3 + 0,7 \cdot 0,15 = \frac{4}{5} \cdot \frac{3}{10} + \frac{7}{10} \cdot \frac{3}{20} = \frac{12}{50} + \frac{21}{200} = \frac{69}{200}$

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b) $P(\text{gratuit} \mid \text{spam}) = 0,6$

$P(\text{gratuit} \mid \text{legitim}) = 0,1$

$$P(\text{spam} \mid \text{oferta} \cap \text{gratuit}) = \frac{P(\text{oferta} \cap \text{gratuit} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{oferta} \cap \text{gratuit})}$$

$$= \frac{P(\text{oferta} \mid \text{spam}) \cdot P(\text{gratuit} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{oferta}) \cdot P(\text{gratuit})}$$

$$P(\text{gratuit}) = \frac{6}{10} \cdot \frac{3}{10} + \frac{1}{10} \cdot \frac{7}{10} = \frac{25}{100} = \frac{1}{4}$$

$$P(\text{spam} \mid \text{oferta} \cap \text{gratuit}) = \frac{\frac{8}{10} \cdot \frac{6}{10} \cdot \frac{3}{10}}{\frac{1}{4} \cdot \frac{69}{200}} = \frac{2448 \cdot 200}{1000 \cdot 23} = \frac{192}{5}$$

$$P(\text{spam} \mid \text{oferta} \cap \text{gratuit}) = \frac{P(\text{oferta} \cap \text{gratuit} \mid \text{spam}) \cdot P(\text{spam})}{P(\text{oferta} \cap \text{gratuit})}$$

$$P(\text{oferta} \cap \text{gratuit} \mid \text{spam}) = P(\text{oferta} \mid \text{spam}) \cdot P(\text{gratuit} \mid \text{spam})$$

$$= 0,6 \cdot 0,8 = 0,48$$

$$P(\text{oferta} \cap \text{gratuit} \mid \text{legitim}) = 0,1 \cdot 0,15 = 0,015$$

$$P(\text{oferta} \cap \text{gratuit}) = \frac{48}{100} \cdot \frac{3}{10} + \frac{15}{1000} \cdot \frac{7}{10} = \frac{1440 + 105}{10000} = \frac{1545}{10000}$$

$$P(\text{oferta} \cap \text{gratuit} \mid \text{spam}) = \frac{\frac{144}{1000}}{\frac{1545}{10000}} = \frac{1440}{1545}$$

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c) Ipoteza independenței poate fi nerealistă deoarece ambele cuvinte aparțin de același domeniu (marketing), ceea ce face prezența unuia dintre ele să mărească și prezența celuilalt.

d) Un email nou care conține un anumit set de cuvinte este clasificat comparând. $P(\text{Spam} \mid \text{Cuvinte})$ și $P(\text{Legitim} \mid \text{Cuvinte})$

Ex. 3/

a) P - studentul a promovat
D - disciplina aleasă
S - programul de studiu ales

a) $P(P \mid D=A, S=1) = \frac{81}{90} = \frac{90}{100}$

$P(P \mid D=B, S=1) = \frac{19}{20} = \frac{95}{100}$

$P(P \mid D=A, S=2) = \frac{1}{10} = \frac{10}{100}$

$P(P \mid D=B, S=2) = \frac{64}{80} = \frac{80}{100}$

b) Pentru ambele programe de studiu disciplina B are o rată de promovare mai mare

c) $P(P \mid D=A) = \frac{81+1}{90+10} = \frac{82}{100}$

$P(P \mid D=B) = \frac{19}{20} + \frac{64}{80} = \frac{19+16}{100} = \frac{35}{100}$

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d) Ordinea se inversează atunci când agregăm datele datorită ponderilor inegale.

$$P(P|D) = \sum_{S} P(P|D, S=s) \cdot P(S=s|D)$$

e) I $P(P|D=A, S=s_i) \leq P(P|D=B, S=s_i) \quad \forall i$

I $P(P|D=A) > P(P|D=B)$

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